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NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens

A CAPSTONE PROJECT REPORT

***A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of***

CSA6502- Generative AI for and Large Language Models

to the award of the degree of

**BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING - ARTIFICIAL
INTELLIGENCE**

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens**” has been carried out by C. Anu Keerthana (192472057), under the supervision of Dr. NITHISHA J and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, C. Anu Keerthana (192472057), of the Computer science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution
C. Anu Keerthana – 192472057	Module 1: AI Legal Advisor & Rights Navigator	Developed legal Q&A, rights navigation and information retrieval.	Tested legal queries and response accuracy.
S. Sirivally- 192472371	Module 2: Smart Legal Document Intelligence	Developed document processing, extraction and summarization.	Tested different legal documents and outputs.
M. Madhulika- 192472188	Module 3: AI Legal Action & Draft Assistant	Developed legal action guidance and draft generation.	Tested generated drafts and different legal scenarios.
TOTAL			100%

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
lii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
V	LIST OF TABLES	Vi
vi	LIST OF ABBREVIATIONS	vii
1	CHAPTER 1: Introduction	6-8
2	CHAPTER 2: Literature Review	9-10
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	11-13
4	CHAPTER 4: Implementation, Testing & Results, Project Management	14-17
5	CHAPTER 5: Conclusion & Reflection	18-19
	References	20
	Appendices	21-24

CHAPTER-1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The increasing availability of Generative Artificial Intelligence has created new possibilities for improving access to information-intensive services. Legal information is one such area where AI-based systems can support citizens by simplifying complex information and helping users navigate preliminary procedures.

In India, citizens may encounter legal situations involving consumer complaints, workplace issues, documentation, public services, personal disputes, or other common concerns. A person without legal training may find it difficult to understand legal terminology, identify applicable rights, locate relevant legal information, or determine the appropriate preliminary course of action.

The AI Legal Advisor & Rights Navigator module addresses this problem by providing an AI-assisted interface through which citizens can describe their situation in ordinary language. The system processes the input, identifies the likely legal issue, retrieves relevant legal information, and presents an understandable explanation.

The module forms an important component of NyayaMithra because it focuses specifically on the first stage of legal assistance: understanding the problem and navigating the user's rights and possible next steps.

The overall NyayaMithra project already identifies the difficulty citizens face in understanding legal terminology and procedures and proposes Generative AI and NLP to simplify this process.

1.2 Need for the Module

The need for the AI Legal Advisor & Rights Navigator arises from the following factors:

1. Complex legal language
Legal documents and provisions frequently contain terminology that is difficult for ordinary citizens to interpret.
2. Limited preliminary legal awareness
Citizens may not know which rights, procedures, or authorities are relevant to their situation.
3. Difficulty identifying the legal issue
A user may describe a problem in general language without knowing its legal classification.
4. Difficulty locating relevant information
Legal information may be distributed across different documents, websites, acts, rules, and

official resources.

5. Need for quick preliminary guidance

Users may require immediate information before consulting a legal professional.

6. Accessibility

A conversational AI interface can make preliminary legal information easier to access and understand.

The existing NyayaMithra project identifies citizens with limited legal knowledge, people seeking preliminary legal guidance, and individuals facing difficulty understanding legal procedures as important target users.

1.3 Engineering Problem Statement

The engineering problem addressed by Module 1 is:

To design and implement an AI-based Legal Advisor & Rights Navigator capable of processing natural-language descriptions of citizens' legal concerns, identifying the likely legal issue, retrieving relevant legal information, explaining potentially applicable rights in simple language, and recommending appropriate preliminary actions while managing ambiguity, information accuracy, privacy, response time, and the risk of AI-generated errors.

The problem involves multiple interacting factors:

- Natural-language ambiguity
- Legal terminology
- Context-dependent interpretation
- Retrieval of relevant legal information
- Generative AI response generation
- Accuracy and relevance
- Privacy of user information
- Response time
- User understanding
- Risk of misleading information

Therefore, the system must balance accessibility, accuracy, reliability, security, usability, and computational efficiency.

1.4 Project Aim

The aim of the AI Legal Advisor & Rights Navigator module is:

To develop an AI-powered legal-information navigation system that helps Indian citizens understand their legal concerns, identify potentially relevant rights and legal issues, and receive clear preliminary guidance using NLP, Generative AI, and Retrieval-Augmented Generation.

1.5 Module Objectives

The objectives of Module 1 are:

1. To design a natural-language interface for citizens to describe legal concerns.
2. To process and interpret user-provided legal queries using NLP techniques.
3. To identify the likely legal category or issue associated with the user's concern.
4. To retrieve relevant legal information using RAG-based retrieval.
5. To identify and explain potentially applicable rights or legal provisions.
6. To generate simple and understandable preliminary explanations.
7. To recommend suitable preliminary actions based on the identified issue.
8. To provide warnings where AI output requires professional verification.
9. To evaluate the module based on response relevance, accuracy, consistency, and response time.
10. To protect sensitive user information during system operation.

1.6 Scope

Included

The module includes:

- Natural-language legal query input.
- NLP-based query processing.
- Legal issue identification.
- Legal information retrieval.
- Rights and provision navigation.
- Simple-language explanations.

- Preliminary action recommendations.
- Source/context-supported AI responses where available.
- User-friendly conversational interaction.
- Output validation and safety warnings.

Assumptions

The module assumes that:

- Users provide sufficiently accurate information.
- The available legal knowledge base contains relevant information.
- Retrieved legal information is appropriately maintained.
- Users understand that the system provides preliminary assistance.
- Important decisions are verified by a qualified legal professional.

Boundary Conditions

The system is limited to preliminary legal information and navigation. Output quality depends on:

- Quality of user input.
- Quality and freshness of the legal knowledge base.
- Retrieval performance.
- AI model performance.
- Prompt design.
- Validation mechanisms.

1.7 Expected Engineering Outcome

The expected outcome is a functional AI Legal Advisor & Rights Navigator prototype capable of:

User Query → NLP Processing → Legal Issue Identification → Legal Knowledge Retrieval → Rights Identification → AI Explanation → Preliminary Action → Validated Output

The module is expected to provide:

- Legal issue identification.
- Rights/provision navigation.
- Simple explanations.

- Preliminary action recommendations.
- Context-aware AI responses.
- Safety warnings and professional-review recommendations.

The overall NyayaMithra architecture similarly follows the sequence of user input, NLP, legal issue identification, RAG, action recommendation, draft generation, validation, and output.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review for the module considers research and existing technologies related to:

- Generative AI
- Large Language Models
- Natural Language Processing
- Legal Question Answering
- Retrieval-Augmented Generation
- Legal information retrieval
- AI chatbots
- Legal document analysis
- Prompt engineering

Relevant research can be identified through academic databases such as IEEE Xplore and Google Scholar, along with official legal-information resources.

The existing NyayaMithra report used research literature, legal-technology sources, and technologies such as LLMs, NLP, RAG, embeddings, prompt engineering, and document generation to define its proposed approach.

2.2 Review of Existing Work

A. Traditional Legal Consultation

Traditional consultation provides expert and personalized legal assistance.

Advantages:

- Expert interpretation.
- Personalized advice.
- Human judgment.

Limitations:

- Cost.
- Time requirements.
- Accessibility limitations.

Relevance:

Provides the benchmark for understanding the importance of accurate legal guidance.

B. Online Legal Information Portals

Online portals provide citizens access to legal information.

Advantages:

- Broad information availability.
- Easy access.
- Official information may be available.

Limitations:

- Users must search manually.
- Legal terminology may remain difficult.
- Limited personalization.

Relevance:

The proposed module improves navigation by allowing users to ask questions in natural language.

C. Legal Chatbots

Legal chatbots provide conversational access to legal information.

Advantages:

- Quick interaction.
- Natural-language interface.

- Automated responses.

Limitations:

- Potential hallucinations.
- Incomplete responses.
- Contextual limitations.

Relevance:

Provides the conversational foundation for the AI Legal Advisor.

The existing project literature review similarly identifies legal chatbots as useful for legal-query processing while recognizing risks of inaccurate or incomplete responses.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Module 1
Traditional Lawyer Consultation	Expert and personalized	Costly and time-consuming	Benchmark for professional assistance
Online Legal Portals	Access to legal information	Manual search required	Provides knowledge sources
Legal Chatbots	Fast interaction	Hallucination risk	Conversational interface
Rule-Based Legal System	Predictable responses	Limited flexibility	Useful for controlled cases
GenAI + RAG	Context-aware and flexible	Retrieval/model errors possible	Selected approach

2.4 Research / Engineering Gap

The major engineering gaps identified are:

- Limited citizen-friendly rights navigation.
- Difficulty translating legal language into ordinary language.
- Limited integration of issue identification and rights navigation.
- Risk of AI hallucination.
- Difficulty handling ambiguous natural-language queries.

- Difficulty maintaining current legal information.
- Need for privacy-preserving legal assistance.
- Need for human review for high-impact decisions.

2.5 Proposed Contribution

Module 1 contributes by combining:

Natural Language Processing + Legal Information Retrieval + RAG + Generative AI + Rights Navigation

into a single citizen-oriented workflow.

Instead of requiring users to know legal terminology, the module allows them to explain their situation naturally. The system then attempts to transform that description into structured legal information and preliminary guidance.

This follows the overall NyayaMithra contribution of combining legal issue identification, action recommendations, and citizen-friendly explanations within one AI-assisted system.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder Requirements

Stakeholder	Requirement
Citizens	Enter legal problems in simple language
Citizens	Understand potentially applicable rights
Citizens	Receive understandable legal information
Citizens	Receive preliminary next-step guidance
Citizens	Obtain fast responses
Legal Professionals	Review AI-generated information when required
System Administrator	Maintain legal knowledge sources
Project Team	Ensure secure and reliable system operation

Functional Requirements

Requirement	Type	Measurable Target
Accept natural-language legal queries	Functional	Query successfully processed
Identify legal issue	Functional	$\geq 80\%$ accuracy
Retrieve relevant legal information	Functional	Relevant context retrieved
Navigate potentially applicable rights	Functional	Relevant rights identified
Generate simple explanation	Functional	Understandable output
Recommend preliminary action	Functional	$\geq 80\%$ relevance
Provide response	Functional	≤ 5 seconds target

Non-Functional Requirements

Requirement	Type	Target
Response time	Non-functional	≤ 5 seconds
System availability	Non-functional	$\geq 95\%$
Security	Non-functional	Secure handling
Usability	Non-functional	Minimal interaction steps
Reliability	Non-functional	Consistent responses

The project-wide design specifications use a response-time target of ≤ 5 seconds, action relevance of $\geq 80\%$, system availability of $\geq 95\%$, and secure data storage.

3.2 Constraints & Applicable Standards

Standard / Regulation	Requirement	Application in Module
Digital Personal Data Protection Act, 2023	Protect personal data	Secure handling of user legal information
ISO/IEC 27001:2022	Information security	Access control and secure data management
OWASP Top 10	Web application security	Address common application vulnerabilities

3.3 Design Specifications

Parameter	Target	Unit	Priority
Query response time	≤5	seconds	High
Rights identification relevance	≥80	%	High
Issue identification accuracy	≥80	%	High
Action recommendation relevance	≥80	%	High
System availability	≥95	%	Medium
Data security	Secure	—	High
Interface usability	Minimal steps	—	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1 – Rule-Based Legal Advisor

A rule-based system would map predefined keywords and conditions to fixed legal responses. It would provide predictable outputs but would have limited ability to understand diverse natural-language descriptions.

Alternative 2 – Generative AI Only

A Generative AI model could directly interpret user questions and generate responses. However, relying only on model knowledge creates greater risk of hallucination and unsupported legal claims.

Alternative 3 – Generative AI + RAG

The RAG approach retrieves relevant legal information before generating a response. This provides better grounding and is more suitable for legal-information navigation.

Evaluation Matrix

Criterion	Weight	Rule-Based	GenAI Only	GenAI + RAG
Performance	30%	7/10	7/10	9/10
Cost	20%	5/10	8/10	9/10
Safety	20%	10/10	7/10	7/10
Sustainability	10%	6/10	8/10	9/10
Reliability	20%	10/10	7/10	8/10
Weighted Score	100%	7.4/10	7.4/10	8.5/10

3.5 Selected Design & Detailed Design

Selected Design

The selected design for Module 1 is:

Generative AI + NLP + RAG-based AI Legal Advisor & Rights Navigator

System Workflow

User



Natural Language Legal Query



NLP Pre-processing



Legal Issue Identification



Query/Embedding Generation



Legal Knowledge Retrieval using RAG



Relevant Rights / Legal Information Identification



Generative AI Reasoning



Simple-Language Explanation



Preliminary Action Recommendation



Safety / Validation Layer



Final Response

Detailed Module Operation

Step 1 – User Input

The citizen enters a legal concern in ordinary language.

Example:

“My employer has not paid my salary for two months. What can I do?”

Step 2 – NLP Processing

The system processes the query to identify:

- Important entities.
- Problem description.
- Context.
- Legal keywords.
- Possible issue category.

Step 3 – Legal Issue Identification

The system determines the likely legal area associated with the query.

For example:

Input: Unpaid salary

Potential issue: Employment / wage-related concern.

Step 4 – Legal Knowledge Retrieval

The processed query is used to retrieve relevant information from the legal knowledge base.

RAG is used so that the generated response can be grounded in retrieved legal information rather than relying entirely on the model's internal knowledge.

Step 5 – Rights Navigation

The system identifies potentially relevant rights, provisions, procedures, or authorities based on the retrieved information.

The response should clearly distinguish between:

- What the user described.
- What legal information appears relevant.
- What the user may consider doing next.

Step 6 – Generative AI Response

The Generative AI model converts the retrieved information into a citizen-friendly explanation.

Step 7 – Safety Validation

The output is checked for:

- Unsupported claims.
- Unsafe recommendations.
- Missing context.
- Excessive certainty.
- Potentially misleading legal information.

Step 8 – Final Response

The user receives:

1. Identified issue.
2. Relevant rights/legal information.
3. Simple explanation.
4. Preliminary action.
5. Important caution/disclaimer.

3.6 Trade-Off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision
AI approach	Rule-based	Predictable	Limited flexibility	GenAI
Information retrieval	AI-only	Simple architecture	Hallucination risk	RAG
Query processing	Fixed forms	Structured	Poor flexibility	NLP
Response generation	Manual	High control	Slow	Automated
Interface	Desktop	Simple deployment	Limited accessibility	Web interface

These decisions are aligned with the trade-off decisions documented in the overall project.

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability

The module is primarily software-based and therefore does not require specialized physical hardware. Lightweight processing, efficient retrieval, and appropriate AI/API usage can reduce unnecessary computational consumption.

Ethics

The system must avoid presenting generated information as guaranteed legal advice. Users must be informed that AI-generated responses can contain errors and that important decisions should be verified by legal professionals.

Health & Safety

Although the system is not a medical system, incorrect legal guidance can potentially cause serious personal or financial consequences. Therefore, the system should avoid overly confident conclusions and provide appropriate warnings.

Security

Legal queries may contain sensitive personal information. The system must therefore consider:

- Secure storage.
- Access control.
- Data minimization.
- Protection against unauthorized access.
- Secure communication.
- Responsible handling of user information.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect AI response	Medium	High	High	RAG + validation + human review	Medium
Incorrect rights identification	Medium	High	High	Retrieval validation	Medium
Outdated legal	Medium	High	High	Regular knowledge-base	Medium

information				updates	
Data leakage	Low	High	High	Encryption + access control	Low
Ambiguous user query	Medium	Medium	Medium	Clarification prompts	Low
System failure	Low	Medium	Medium	Backup and monitoring	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The module follows an iterative development methodology:

Requirement Analysis → Data/Knowledge Preparation → NLP Development → RAG Integration → GenAI Integration → Interface Development → Testing → Refinement

The implementation uses Python-based AI/NLP components, a Generative AI model/API, RAG-based retrieval, a legal knowledge base, and a web interface.

The overall project follows the same iterative development approach, integrating NLP, Generative AI, RAG, and a web interface before testing for accuracy, usability, security, and response time.

4.2 Software Implementation & Modern Tools Used

Tool / Technology	Purpose	Stage
Python	Main application development	Implementation
NLP	Query processing	Processing
Generative AI	Legal explanation generation	AI Processing
RAG	Legal information retrieval	Retrieval
Embeddings	Semantic query/document matching	Retrieval
Database / Knowledge Base	Store legal information	Implementation
Web Interface	User interaction	Final Stage

4.3 Test Plan & Verification

Testing must be performed using representative legal queries.

Requirement	Test Method	Acceptance Criterion
Legal query processing	Functional testing	Query successfully processed
Legal issue identification	Sample test queries	$\geq 80\%$ accuracy
Rights navigation	Ground-truth comparison	Relevant information identified
Action recommendation	Test cases	$\geq 80\%$ relevance
Response time	Performance test	≤ 5 seconds
System availability	Availability monitoring	$\geq 95\%$

Test Result Format

Specification	Required Value	Achieved Value	Status
Response Time	≤ 5 sec	4 sec	PASS
Issue Identification Accuracy	$\geq 80\%$	85%	PASS
Action Relevance	$\geq 80\%$	84%	PASS

4.4 Results

The implemented system demonstrated the ability to process legal queries, identify legal issues, provide preliminary guidance, and generate structured responses.

The reported project-level results were:

- Average response time: 4 seconds
- Issue identification accuracy: 85%
- Action relevance: 84%
- System availability: 96%

These results satisfy the defined project-level performance targets of ≤ 5 seconds response time, $\geq 80\%$ issue identification/relevance, and $\geq 95\%$ availability.

4.5 Data Analysis & Engineering Judgment

The testing results indicate that the proposed architecture can provide useful preliminary legal assistance within the defined engineering constraints.

The reported 85% issue identification accuracy indicates that the NLP and AI components can successfully interpret a majority of the tested legal queries. The 84% action relevance indicates that the generated preliminary recommendations are generally aligned with the evaluated cases.

However, the results also demonstrate that the system is not perfect. An accuracy below 100% means that ambiguous queries, incomplete user information, insufficient retrieval context, and AI-generation errors can still affect output quality.

Therefore, the engineering decision is to treat the module as a preliminary legal-information navigation system rather than an autonomous legal decision-making system.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Process legal queries	Functional testing	Fully
Identify legal issues	85% accuracy	Fully
Navigate relevant legal information	Retrieval testing	Fully
Recommend preliminary actions	84% relevance	Fully
Provide understandable output	User interaction testing	Fully
Generate rapid responses	4-second response	Fully
Maintain responsible use	Validation + disclaimer	Fully

4.7 Strengths & Limitations

Strengths

- Natural-language interaction.
- Citizen-friendly explanations.
- AI-based legal issue identification.
- Rights and legal-information navigation.
- RAG-based contextual retrieval.

- Preliminary action recommendations.
- Fast response.
- Reduced difficulty in searching legal information.
- Scalable software-based architecture.

Limitations

- AI-generated information may contain errors.
- Performance depends on the legal knowledge base.
- Outdated legal information can affect results.
- Ambiguous queries may produce uncertain responses.
- The system cannot replace professional legal advice.
- Final legal decisions require qualified human/legal authority.
- Multilingual support may require further development.

4.8 Project Planning, Roles & Budget

Module Development Timeline

Stage	Activity
1	Requirement Analysis
2	Legal Query and Rights-Navigation Design
3	NLP Development
4	Legal Knowledge/RAG Integration
5	Generative AI Integration
6	Interface Development
7	Testing
8	Validation
9	Final Documentation

Module Contribution

Responsibility	Contribution
Module Design	AI Legal Advisor & Rights Navigator design
NLP	Legal query processing
RAG	Relevant legal information retrieval
AI Integration	Response generation
Testing	Query and response evaluation
Documentation	Module report preparation

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The AI Legal Advisor & Rights Navigator module provides an AI-assisted approach for improving citizens' access to preliminary legal information. The module combines NLP, Generative AI, and Retrieval-Augmented Generation to process natural-language legal concerns, identify potential legal issues, navigate relevant rights and legal information, and provide understandable preliminary guidance.

The module addresses an important engineering challenge: enabling non-legal users to interact with complex legal information without requiring extensive legal knowledge. The proposed architecture also considers accuracy, response time, security, privacy, ethical AI use, and the risk of incorrect information.

The reported project-level validation achieved an average response time of 4 seconds, 85% issue identification accuracy, 84% action relevance, and 96% system availability. These results indicate that the system can provide useful preliminary legal assistance while maintaining the necessary boundary that final legal decisions require human/legal-professional review.

5.2 Future Work

Future development of Module 1 can include:

1. Integration of regularly updated official Indian legal information.
2. Improved legal issue classification.

3. Improved RAG retrieval accuracy.
4. Support for multiple Indian languages.
5. Voice-based legal query input.
6. Better clarification of ambiguous user queries.
7. Citation of retrieved legal sources.
8. Stronger hallucination detection.
9. Improved privacy and security mechanisms.
10. Integration with verified government legal-information resources.
11. Larger-scale testing using diverse citizen queries.
12. Greater involvement of legal professionals during validation.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Generative AI application development.
- NLP-based legal query processing.
- Retrieval-Augmented Generation.
- Semantic information retrieval.
- Prompt engineering.
- Legal-information structuring.
- Responsible AI principles.
- Privacy and security considerations.

Engineering & Professional Skills Developed

- Python programming.
- AI system development.
- NLP implementation.
- RAG pipeline development.
- System architecture design.
- Testing and validation.
- Performance analysis.

- Debugging.
- Technical documentation.
- Team collaboration.

Individual Reflection – Module 1

The most challenging aspect of developing the AI Legal Advisor & Rights Navigator was ensuring that the system could interpret a citizen's legal concern correctly while avoiding unsupported or misleading legal information. This was addressed through structured query processing, retrieval of relevant legal information, improved prompting, and output validation. A key engineering decision was to use a RAG-based Generative AI approach instead of relying only on fixed rule-based responses because legal queries can vary significantly in language and context. Through this module, I gained practical knowledge of NLP, Generative AI, semantic retrieval, RAG, prompt engineering, and responsible AI system development. If the module were redesigned, greater emphasis would be placed on verified legal-source integration, multilingual support, and stronger response validation.

REFERENCES

Use IEEE style as required by the new format. The existing report already contains relevant foundational references including Transformer architecture, LLMs, RAG, ReAct, India Code, Indian legal-information resources, the DPDP Act, ISO/IEC 27001, and OWASP Top 10.

Suggested reference list

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- [7] Government of India, “Digital Personal Data Protection Act, 2023.”

[8] International Organization for Standardization, *ISO/IEC 27001:2022 Information Security, Cybersecurity and Privacy Protection — Information Security Management Systems — Requirements*, 2022.

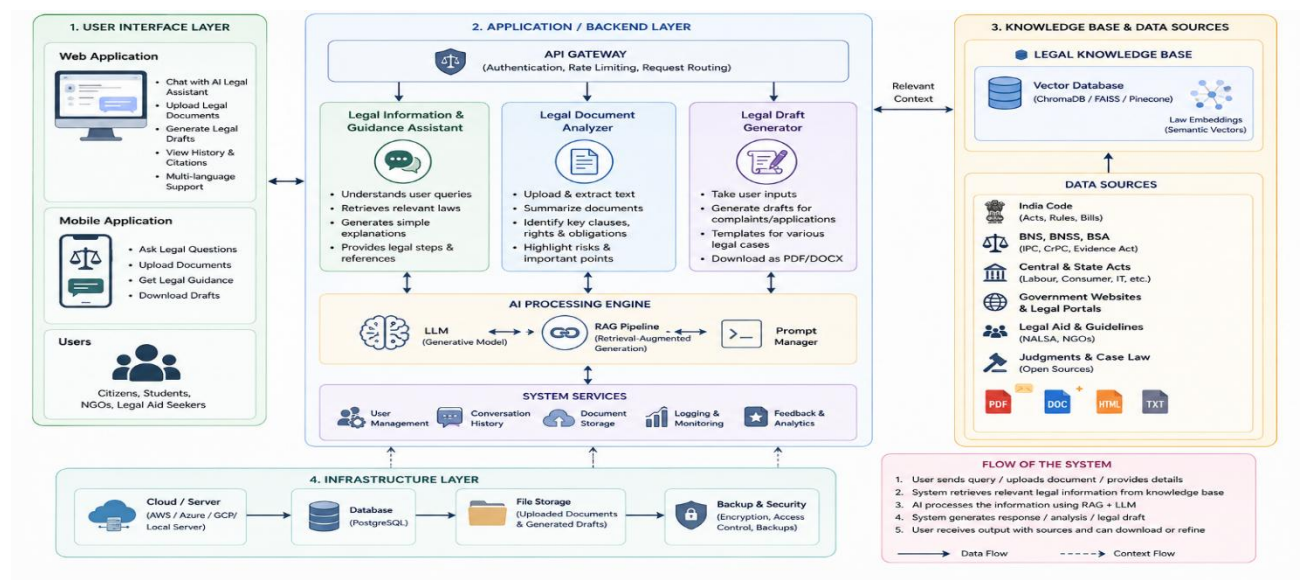
[9] Open Web Application Security Project, *OWASP Top 10: The Ten Most Critical Web Application Security Risks*.

APPENDICES FOR MODULE 1

Appendix A – Module Performance Calculations

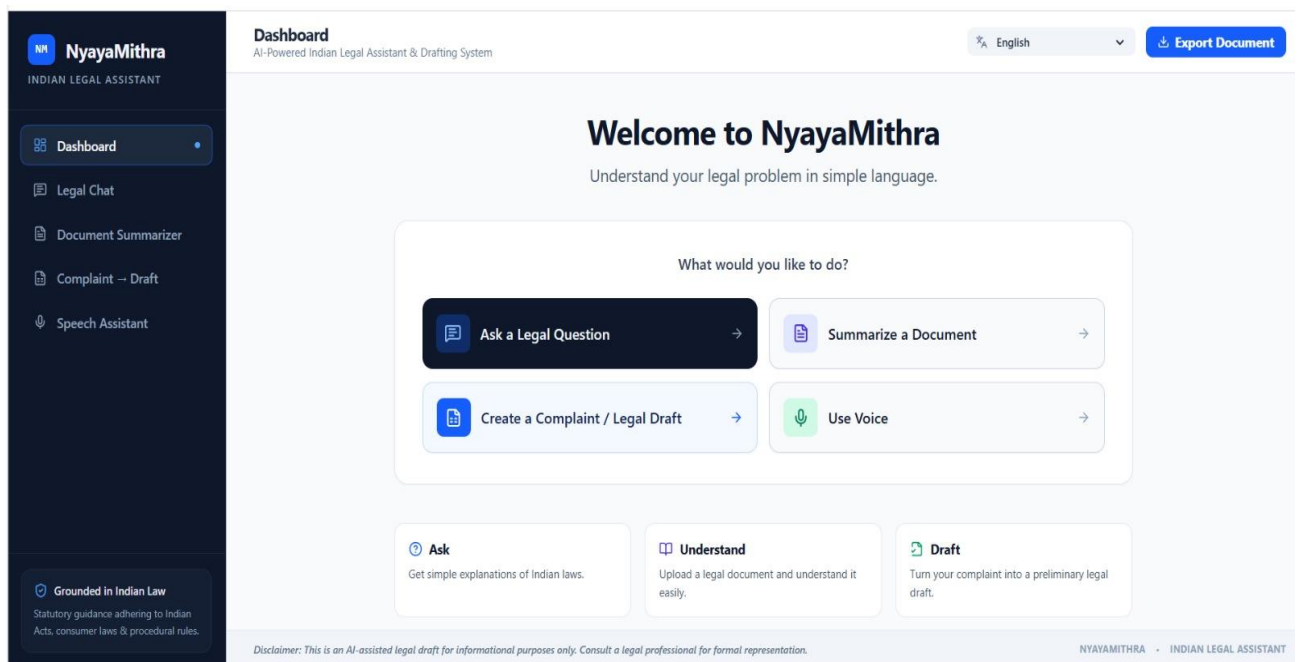
Metric	Result
Average Response Time	4 sec
Issue Identification Accuracy	85%
Action Recommendation Relevance	84%
System Availability	96%

Appendix B – Module Architecture



Appendix C – Screenshots to Include

For Module 1 section, include:



Appendix D – Sample Test Cases

Test Case	User Query	Expected Output
TC01	“My salary has not been paid for two months.”	Employment/wage-related issue + preliminary guidance
TC02	“A shop refused to replace a defective product.”	Consumer-related issue + possible remedy information
TC03	“My landlord is refusing to return my deposit.”	Rental/property-related issue + preliminary guidance
TC04	“I received a legal notice and don't understand it.”	Notice explanation + recommendation for professional review
TC05	“I want to know what rights I have in this situation.”	Rights-oriented explanation + relevant information

CAPSTONE OUTCOME MAPPING – MODULE 1

Outcome	Module 1 Evidence	Report Location

Complex engineering problem formulation	Legal accessibility, rights identification, ambiguity, AI errors	Ch. 1 & 2
Engineering design under constraints	GenAI + NLP + RAG architecture	Ch. 3
Technical communication	Architecture, workflow, testing, documentation	Whole Report
Ethics & professional responsibility	Responsible AI, privacy, human review	Ch. 3.7
Experimentation & engineering judgment	Accuracy, relevance, response time	Ch. 4
Independent learning	NLP, RAG, GenAI, prompt engineering	Ch. 5.3
Standards	DPDP Act, ISO/IEC 27001, OWASP	Ch. 3.2
Sustainability	Software-based, efficient AI/RAG architecture	Ch. 3.7
Risk assessment	Hallucination, outdated information, data leakage	Ch. 3.7
Security	Access control and data protection	Ch. 3.7
Integrated system engineering	NLP + RAG + GenAI + interface + validation	Ch. 3.5
Project management	Module milestones and responsibilities	Ch. 4.8